

Statistics for Data Science Day 4: Measures of Dispersion

Beyond the Average: Five Measures of Spread Every Data Scientist Must Know



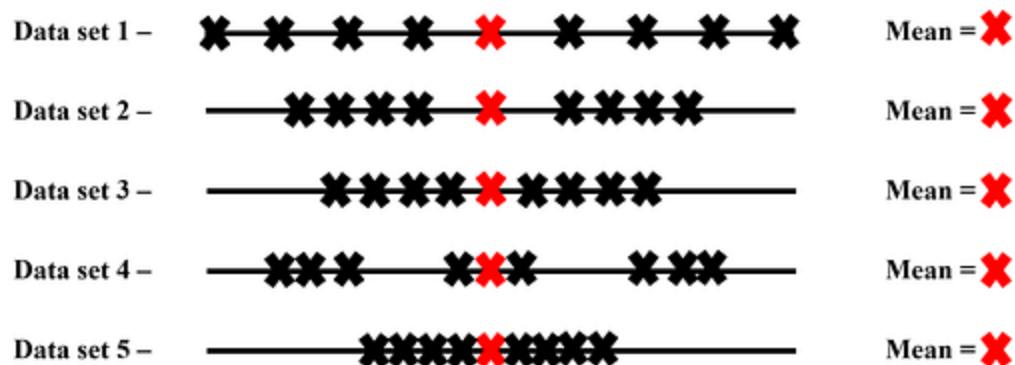
DATA SCIENCE WITH SHIKHAR

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In Day 3, we learned how mean, median, and mode summarise the centre of a dataset. But knowing the centre alone is like knowing the average temperature of a city, it tells you very little about whether you'll need a jacket or sunscreen on any given day. Today we tackle the other half of the picture: how spread out is the data?

Why the Centre Isn't Enough

Consider these five datasets, they all have the **same mean**, yet they are clearly very different:



Dataset 5 has very little variability. Dataset 1 has very high variability. If you only reported the mean, all five would look identical. **Measures of**

dispersion are what distinguish them.

In this article we cover five key measures of spread:

- **Range**
 - **Variance**
 - **Standard Deviation**
 - **Median Absolute Deviation (MAD)**
 - **Interquartile Range (IQR)**
-

Range

Range is the simplest measure of spread which is just the difference between the maximum and minimum values in a dataset.

Range=Maximum–Minimum

Example

Salaries (in USD) of 10 data scientists at an organisation:

[78.3k, 72k, 75k, 80k, 73.5k, 70k, 78k, 85k, 71k, 73k]

Range = \$85,000 – \$70,000 = \$15,000

This tells us the salaries span a \$15,000 gap, a useful first impression.

Limitations of Range

- **Ignores all middle values** : only two data points (max and min) determine it entirely

- **Extremely sensitive to outliers** : one extreme value distorts it completely

For example, replace the \$85k salary with \$300k:

[78.3k, 72k, 75k, 80k, 73.5k, 70k, 78k, 71k, 73k, 300k]

| Range = \$300,000 – \$70,000 = \$230,000

The range jumps from \$15k to \$230k because of a single outlier despite the other nine salaries barely changing. This is why we need more robust measures.

Variance

Variance measures the **average squared deviation of each data point from the mean**. Unlike range, it considers every value in the dataset, making it a far more comprehensive measure of spread.

Population vs. Sample Variance

There are two formulas depending on whether your data is the entire population or just a sample:

Population Variance:

$$\sigma^2 = \frac{\sum_{i=1}^N (X_i - \mu)^2}{N}$$

Sample Variance:

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

The key difference is dividing by **n-1** instead of **n** for sample variance. This adjustment is called **Bessel's correction**, it compensates for the fact that a sample tends to underestimate the true population spread. We'll cover this in detail later.

Worked Example

Dataset: {46, 69, 32, 60, 52, 41}

Step 1. Find the Mean:

$$\bar{x} = \frac{46 + 69 + 32 + 60 + 52 + 41}{6} = \frac{300}{6} = 50$$

Step 2. Find deviations from the mean and square them:

Data Point	Deviation ($x_i - \bar{x}$)	Squared Deviation
46	-4	16
69	19	361
32	-18	324
60	10	100
52	2	4
41	-9	81
Total		886

Step 3. Calculate Sample Variance (dividing by $n-1 = 5$):

$$s^2 = \frac{886}{5} = 177.2$$

Why We Square the Deviations

Without squaring, positive and negative deviations would cancel each other out, and the sum would always be zero (or close to it) hence giving no useful information. Squaring ensures all deviations contribute positively to the total spread.

Limitation of Variance

The result is in **squared units**, if your data is in dollars, variance is in “dollars squared.” This makes it hard to interpret directly. That’s precisely why we take the square root and get standard deviation.

Standard Deviation

Standard deviation is simply the **square root of variance**, bringing the measure back into the original units of the data making it directly interpretable alongside the mean.

$$\sigma = \sqrt{\sigma^2} \quad (\text{population}) \qquad s = \sqrt{s^2} \quad (\text{sample})$$

For our dataset {46, 69, 32, 60, 52, 41}:

$$s = \sqrt{177.2} \approx 13.31$$

This means: on average, data points in this set deviate about **13.31 units** from the mean of 50. That's an immediately interpretable statement.

```
import numpy as np
data = [46, 69, 32, 60, 52, 41]

print(np.std(data, ddof=1)) # Sample std dev → 13.31
print(np.var(data, ddof=1)) # Sample variance → 177.2
```

Note: Always use `ddof=1` in NumPy when working with a sample (not the full population). The default `ddof=0` computes population variance/std dev.

Standard Deviation in Practice

Use Case	How SD Helps
Understanding Distribution	High SD = widely spread data; Low SD = tightly clustered data
Finance & Risk	SD of stock returns measures investment volatility
Quality Control	Manufacturing uses SD to monitor consistency of products
Machine Learning	Feature scaling (standardisation) uses mean and SD
Outlier Detection	Values more than 2–3 SDs from the mean are flagged as outliers

The Limitation: Sensitivity to Outliers

Both variance and standard deviation use the **mean** in their calculation. Since mean is itself sensitive to outliers (as we saw in Day 3), variance and SD inherit that same weakness. A single extreme value inflates both significantly.

This is why we need the next two measures.

Median Absolute Deviation (MAD)

MAD is a **robust alternative to standard deviation** for datasets with outliers. Instead of measuring average distance from the *mean*, it measures the median distance from the *median* making it resistant to extreme values.

How to Calculate MAD

Step 1: Find the **Median** of the dataset.

Step 2: Calculate the **absolute deviation** of each data point from the median (i.e., $|x_i - \text{Median}|$).

Step 3: Find the **Median of those absolute deviations**, that's your MAD.

Worked Example

Test scores: {94, 85, 78, 92, 100, 96}

Step 1 — Sort and find Median:

Sorted: [78, 85, 92, 94, 96, 100]

$$\text{Median} = (92 + 94) / 2 = 93$$

Step 2 — Absolute deviations from 93:

Deviation of values in the set above = [15, 8, 1, 1, 3, 7]

Step 3 — Sort absolute deviations and find their median:

Sorted: [1, 1, 3, 7, 8, 15]

$$\text{MAD} = (3 + 7) / 2 = 5$$

The typical score deviates **5 points** from the median, a stable, outlier-resistant result.

When to use MAD over SD: Whenever your dataset contains outliers or is strongly skewed. MAD is commonly used in anomaly detection and robust statistical modelling.

Interquartile Range (IQR)

The IQR is another robust measure of spread that focuses on the **middle 50% of the data**, completely ignoring the top and bottom 25% where outliers tend to live.

Understanding Quartiles First

To compute IQR, you need to understand **quartiles**, the values that divide a sorted dataset into four equal parts:

Quartile	Description	Percentile
Q1 (Lower quartile)	Median of the lower half	25th percentile
Q2 (Median)	Middle of the entire dataset	50th percentile
Q3 (Upper quartile)	Median of the upper half	75th percentile

IQR = Q3 – Q1

Worked Example

17 student test scores: {**98, 94, 94, 62, 73, 87, 77, 68, 72, 85, 99, 100, 86, 60, 52, 100, 80**}

Step 1 — Sort:

[52, 60, 62, 68, 72, 73, 77, 80, 85, 86, 87, 94, 94, 98, 99, 100, 100]

Step 2 — Find Q2 (Median):

17 values in the set → middle value is the 9th → **Q2 = 85**

Step 3 — Find Q1 and Q3:

- Lower half (excluding median): [52, 60, 62, 68, 72, 73, 77, 80] → Q1 = $(68 + 72) / 2 = \mathbf{70}$
- Upper half (excluding median): [86, 87, 94, 94, 98, 99, 100, 100] → Q3 = $(94 + 98) / 2 = \mathbf{96}$

Step 4 — Calculate IQR:

IQR = 96 – 70 = 26

The middle 50% of students scored within a **26-point range**. The outliers (52 and 100) have zero influence on this result.

```
import numpy as np
scores = [52, 60, 62, 68, 72, 73, 77, 80, 85, 86, 87, 94, 94, 98, 99, 100, 100]

Q1 = np.percentile(scores, 25)
Q3 = np.percentile(scores, 75)

IQR = Q3 - Q1

print(f"Q1={Q1}, Q3={Q3}, IQR={IQR}")
```

Note: NumPy's *percentile* function may return slightly different Q1/Q3 values from the manual method above due to interpolation. Both approaches are valid, manual calculation using medians of halves is the standard textbook method, while NumPy uses linear interpolation by default.

IQR and Outlier Detection

IQR has a very common practical use: defining the **fence rule** for detecting outliers in a box plot:

$$\text{Lower Fence} = Q1 - 1.5 \times \text{IQR}$$

$$\text{Upper Fence} = Q3 + 1.5 \times \text{IQR}$$

$$\text{Lower Fence} = 70 - (1.5 \times 26) = 70 - 39 = 31$$

$$\text{Upper Fence} = 96 + (1.5 \times 26) = 96 + 39 = 135$$

Any data point outside these fences is flagged as a potential outlier. Hence any value below 31 or above 135 in the above set, would be

flagged as an outlier. In this dataset, no scores fall outside this range.

This is the exact rule used by box plots in Matplotlib, Seaborn, and virtually every visualisation library.

Spread and Central Tendency Together

A measure of spread only becomes fully meaningful when reported alongside a measure of central tendency. Here's why:

Consider two companies with identical mean salaries of \$96,000:

- **Company A:** [\$20k, \$20k, \$20k, \$20k, \$400k] : SD is huge, mean is misleading
- **Company B:** [\$90k, \$94k, \$96k, \$98k, \$102k] : SD is small, mean is reliable

Same mean, completely different stories. **Always report spread alongside centre.** In practice, the standard convention is:

- For symmetric data: report **Mean $\pm$ Standard Deviation**
 - For skewed data: report **Median and IQR**
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Key Takeaway & Practical Tip

Every time you compute a mean or median in a project, immediately follow it up with a measure of spread. A central tendency without dispersion is an incomplete and potentially misleading summary.

Quick Reference:

- No outliers, symmetric data → **Mean + Standard Deviation**

- Outliers or skewed data → **Median + IQR or MAD**
- Just need a rough, quick range → **Range** (but treat it as a starting point only)

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